

ASSOCIATION ANALYSIS

FREQUENT ITEMSET GENERATION AND RULE GENERATION – PART III

Fabio Stella

Associate Professor

c/o Department of Informatics, Systems and Communication

University of Milano-Bicocca



FREQUENT ITEMSET GENERATION

The following concept will be introduced:

✓ **MAXIMAL FREQUENT ITEMSET**

Each **FREQUENT K-ITEMSET**, Y , can generate up to **$2^k - 2$ ASSOCIATION RULES**, ignoring rules that have empty antecedents or consequents.

An **association rule** can be **extracted** by **partitioning** the itemset Y into two non-empty subsets, X and $Y - X$, such that

$$X \rightarrow Y - X$$

satisfies the confidence threshold.

$Y = \{\text{swiss cheese, cheddar, diet coke}\}$

FREQUENT 3-ITEMSET



$X = \{\text{swiss cheese, cheddar}\}$

$Y - X = \{\text{diet coke}\}$

$\{\text{swiss cheese, cheddar}\} \rightarrow \{\text{diet coke}\}$

All such rules met the support threshold because they are generated from a frequent itemset



In practice, the number of **FREQUENT ITEMSETS** produced from a transaction data set can be very large. It is useful to identify a **small representative set of itemsets** from which **all other frequent itemsets** can be derived.



Two such representations are:

- **MAXIMAL FREQUENT ITEMSETS**
- **CLOSED FREQUENT ITEMSETS**

COMPACT REPRESENTATION OF FREQUENT ITEMSETS

2

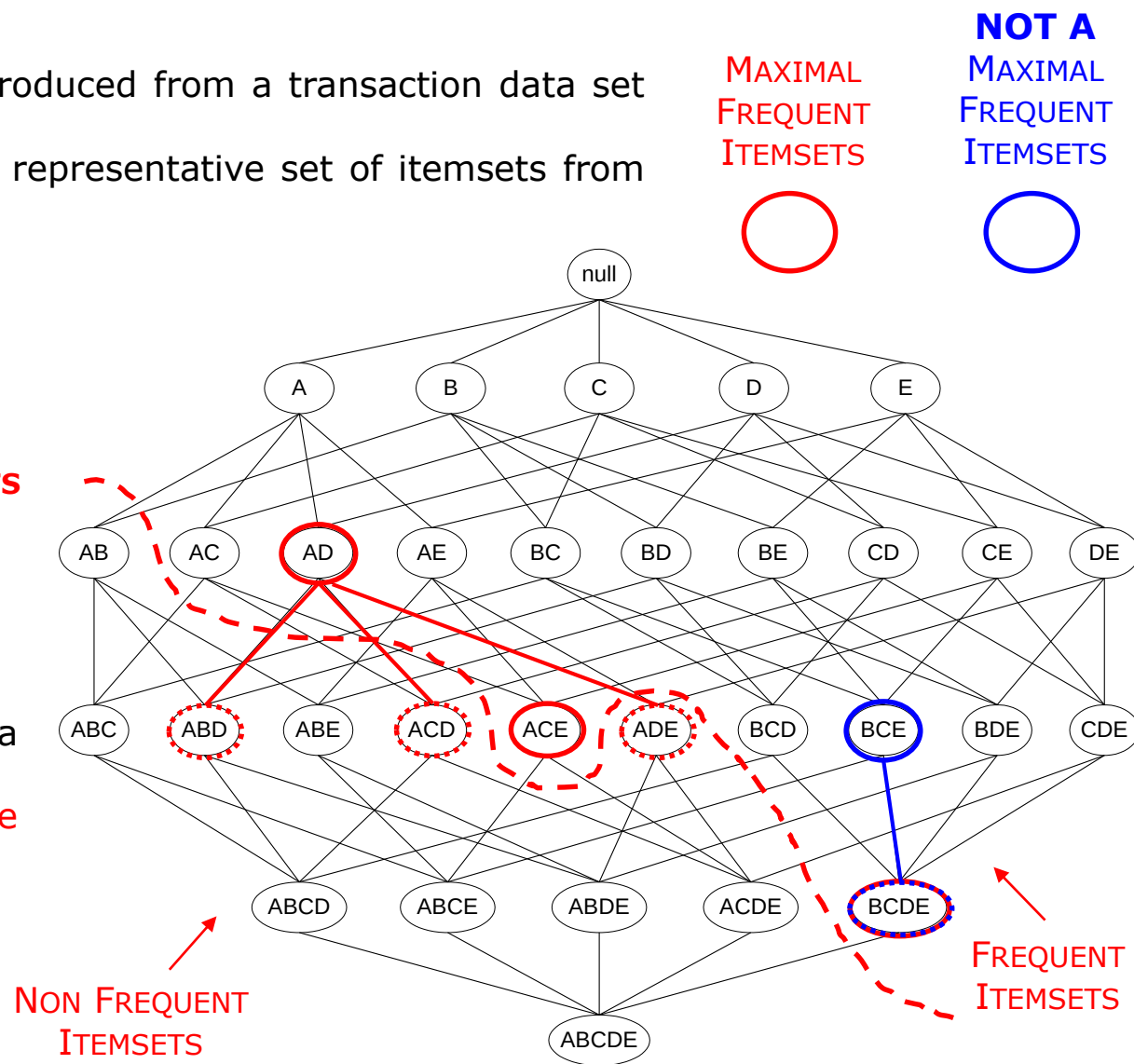
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- **MAXIMAL FREQUENT ITEMSETS**
- **CLOSED FREQUENT ITEMSETS**

A **MAXIMAL FREQUENT ITEMSET** is defined as a Frequent Itemset for which none of its immediate supersets are frequent.



COMPACT REPRESENTATION OF FREQUENT ITEMSETS

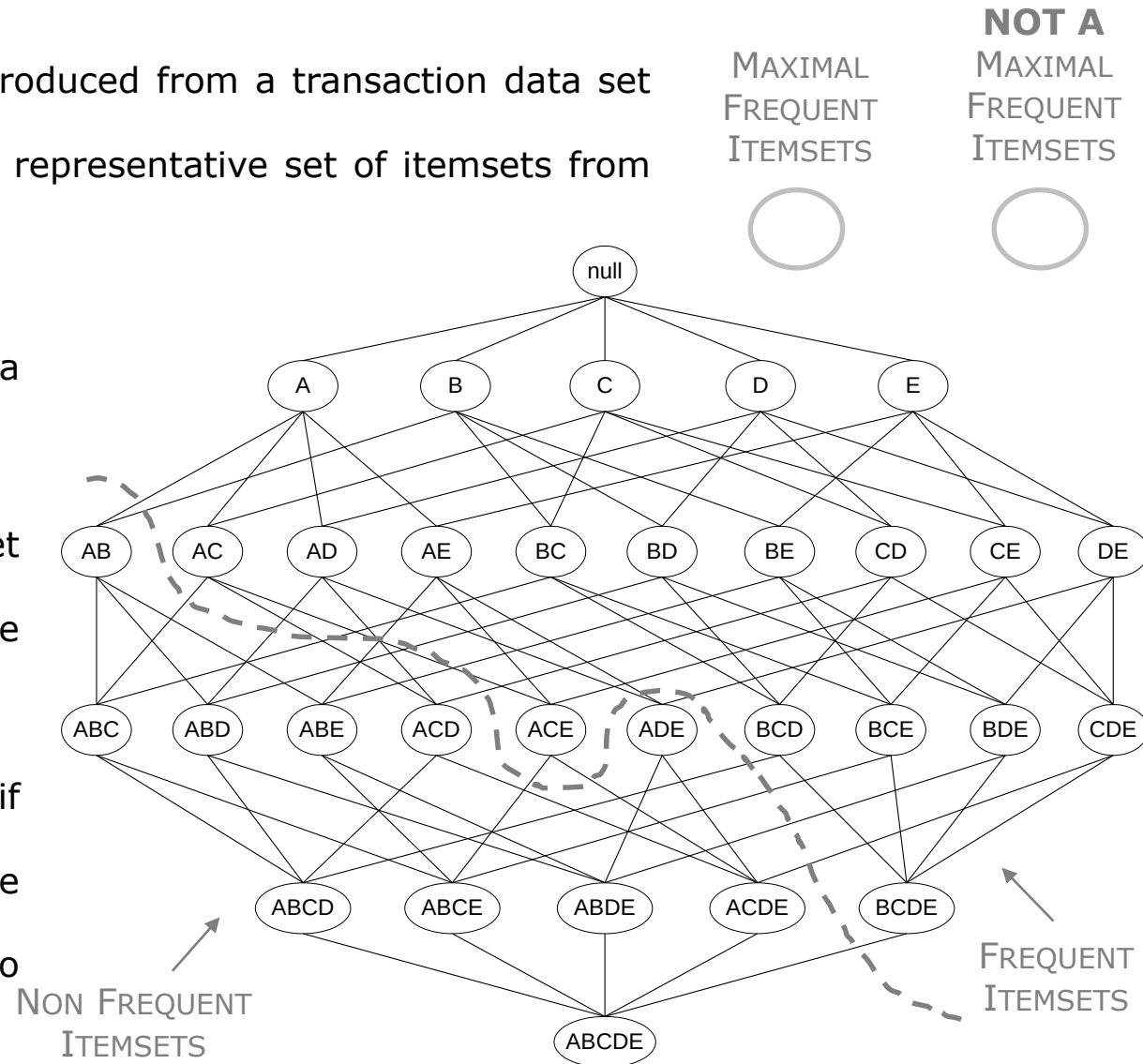
2

In practice, the number of **FREQUENT ITEMSETS** produced from a transaction data set can be very large. It is useful to identify a small representative set of itemsets from which all other frequent itemsets can be derived.

MAXIMAL FREQUENT ITEMSETS effectively provide a compact representation of frequent itemsets.

MAXIMAL FREQUENT ITEMSETS form the smallest set of itemsets from which all frequent itemsets can be derived.

MAXIMAL FREQUENT ITEMSETS are practical only if an efficient algorithm exists to explicitly find the maximal frequent itemsets without having to enumerate all their subsets.



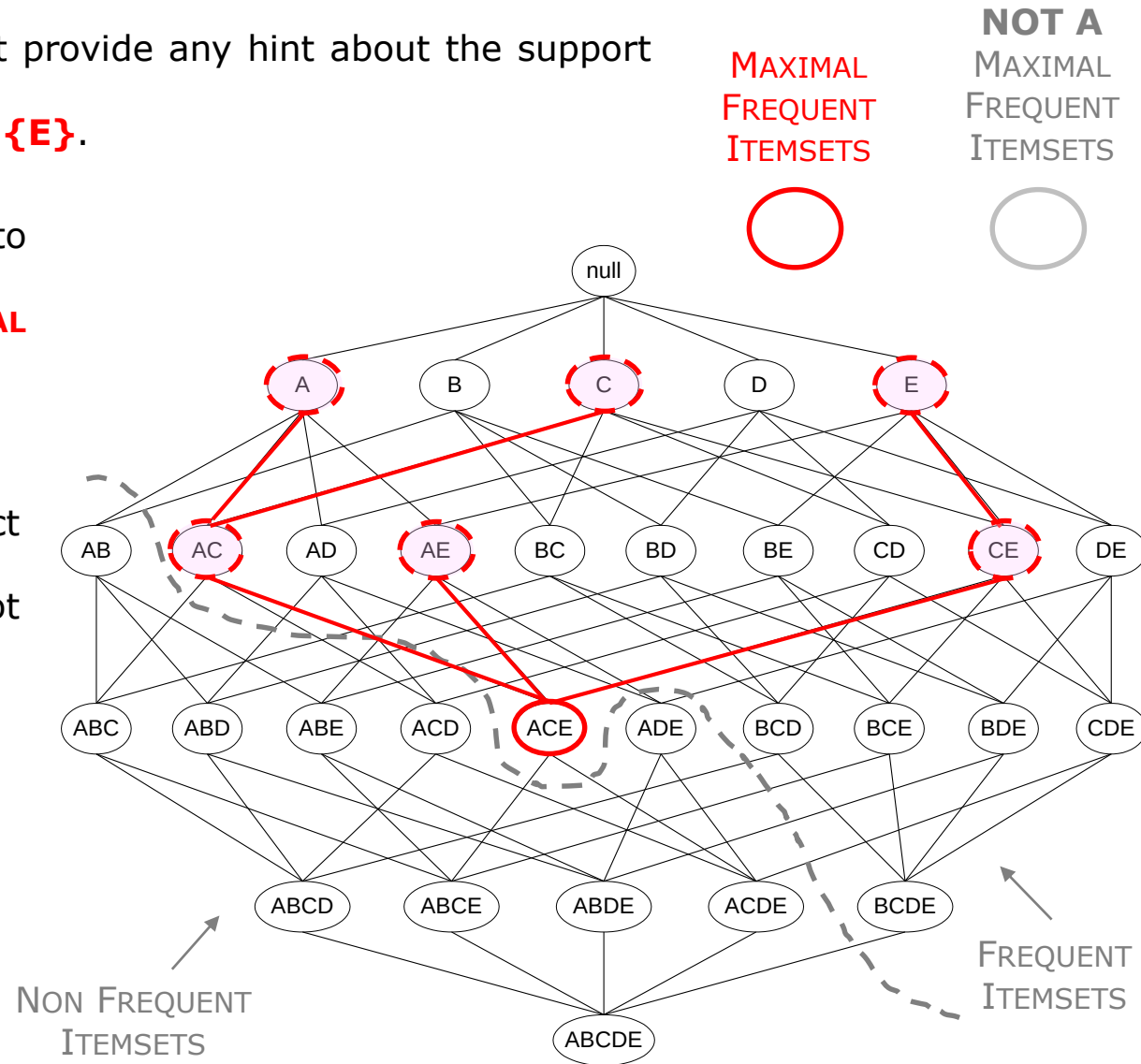
COMPACT REPRESENTATION OF FREQUENT ITEMSETS

2

The **MAXIMAL FREQUENT ITEMSET** {ACE} does not provide any hint about the support of its subsets {AE}, {AC}, {A}, {C}, {CE} and {E}.

An additional pass over the data set is needed to determine the support counts of the **Non-MAXIMAL FREQUENT ITEMSETS**.

MAXIMAL FREQUENT ITEMSETS provide a compact representation of Frequent Itemsets but do not contain the support information of their subsets.



COMPACT REPRESENTATION OF FREQUENT ITEMSETS

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The **MAXIMAL FREQUENT ITEMSET** **{ACE}** does not provide any hint about the support of its subsets **{AE}**, **{AC}**, **{A}**, **{C}**, **{CE}** and **{E}**.

An additional pass over the data set is needed to determine the support counts of the **NON-MAXIMAL FREQUENT ITEMSETS**.

MAXIMAL FREQUENT ITEMSETS provide a compact representation of Frequent Itemsets but do not contain the support information of their subsets.

In some cases, it might be desirable to have a **minimal representation of frequent itemsets that preserves the support information**.

